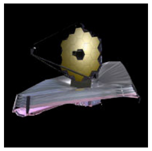

11376 - Water in Little Red Dots? A Pathfinder Program with the MIRI/LRS to Confirm the Cool Temperatures of LRDs

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI/LRS Observations of NEXUS-OBS5-2314				
	3	Obs #3 [2314]	MIRI Low Resolution Spectroscopy	(3) NEXUS-OBS5-2314
MIRI/LRS Observations of MACSJ0647-1045				
	4	Obs #4 [1045]	MIRI Low Resolution Spectroscopy	(4) MACSJ0647-1045

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
MIRI/LRS Observations of GTO-WIDE-UDS14-4446				
	5	Obs #5 [4446]	MIRI Low Resolution Spectroscopy	(5) GTO-WIDE-UDS14-4446

ABSTRACT

Little Red Dots (LRDs) represent one of JWST's most intriguing discoveries. Yet the most basic question about LRDs remain unanswered: Why are LRDs red? Are LRDs red because they are dust-reddened but otherwise normal AGNs, or are they red intrinsically? The latter would be highly unusual for AGNs, as they tend to be intrinsically blue (typical accretion disk temperature ~10^5-10^6 K).

Recently, pronounced water absorption features have been identified in z ~ 2 LRDs (priv. comm.), providing the first evidence that at least some LRDs are indeed cool (T < 4000K); i.e., red intrinsically. However, how common are such cool LRDs? Detecting water absorption requires spectral coverage up to rest-frame ~1.4 microns. This is shifted beyond the NIRSpec/Prism range at z > 3, where the number density of LRDs start to increase.

We propose a MIRI/LRS pathfinder program to reveal the cool temperatures of LRDs. We select a sample of 5 LRDs at z = 3 - 5 that fully span the range in rest-optical continuum shapes, ensuring that the program probes whether molecular absorption is present across cosmic time and under varying physical conditions. While the detection of water absorption provides a model-independent confirmation of the presence of cold gas, preliminary modeling predicts distinct continuum shapes at different temperatures, allowing us to constrain the thermal structure even in the absence of detected absorption features.

This highly efficient program, requiring a very small investment of JWST time, will deliver the first systematic exploration of LRD temperatures and establish a crucial benchmark for understanding their role in early black hole and galaxy evolution.

OBSERVING DESCRIPTION

We are proposing for deep spectroscopic follow-up observations with the MIRI/LRS. The target sample includes five Little Red Dots (LRDs) with numerous existing JWST observations. These existing observations include NIRCam/Imaging (confirming the compact nature of our target sample) in addition to NIRSpec/Spectroscopy (confirming the redshifts, V-shaped spectral energy distributions, and broad-Balmer lines) of these five sources. We utilized the Dawn JWST Archive (DJA) to select the target sample of five LRDs. As described in the scientific justification, these LRDs were selected to span the full range of redshifts, inferred blackbody temperatures, and Balmer-break strengths in order to provide a representative sample of LRDs. The redshift range covered by our sample is between 3.1 and 5.1, spanning nearly one billion years of cosmic time. Our proposed follow-up observations will target rest-infrared wavelengths for these LRDs, including the continuum immediately surrounding the water absorption feature at rest-frame ~1.4 microns.

Our observing strategy is similar to that of other programs with similarly deep MIRI/LRS integrations (e.g., PIDs 3703, 8544, 9165). To quickly summarize this strategy, we adopt the FASTR1 readout pattern with 105-155 groups per integration, 5-19 integrations per exposure, one exposure per dither, and two dithers along the slit nod per visit. Two direct images would be acquired for each of the visits with the MIRI/F560W filter, one for target acquisition (TA) and the other for verification. Verification images are not required but highly recommended for slit spectroscopy of faint targets with the MIRI/LRS to ensure that the target was properly placed within the slit. Self-TA would be performed for the brightest LRD while adopting the FASTGRPAVG16 readout pattern with eight groups per integration and one integration per exposure. TA would be obtained for the fainter LRDs by using bright, nearby stars from Gaia DR3 while adopting the FAST readout pattern with 4-10 groups per integration and one integration per exposure. For each of the targets, verification would adopt the FASTR1 readout pattern with 135 groups per integration and one integration per exposure.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	NEXUS-OBS5-2314	RA: 17 54 10.0550 (268.5418958d) Dec: +65 08 19.71 (65.13881d) Equinox: J2000	Epoch of Position: 2000	

Comments: NEXUS-OBS5-2314 is a Little Red Dot (LRD) at redshift 4.422 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the North Ecliptic Pole (NEP) field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	MACSJ0647-1045	RA: 06 47 44.0174 (101.9334058d) Dec: +70 11 53.76 (70.19827d) Equinox: J2000	Epoch of Position: 2000	

Comments: MACSJ0647-1045 is a Little Red Dot (LRD) at redshift 4.528 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the MACS J0647+7015 (MACSJ0647) lensing cluster field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	GTO-WIDE-UDS14-4446	RA: 02 17 4.9709 (34.2707121d) Dec: -05 13 3.62 (-5.21767d) Equinox: J2000	Epoch of Position: 2000	

Comments: GTO-WIDE-UDS14-4446 is a Little Red Dot (LRD) at redshift 4.683 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the CANDELS Ultra Deep Survey (UDS) field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	TA___NEXUS-OBS5-2314	RA: 17 54 8.5637 (268.5356821d) Dec: +65 07 44.70 (65.12908d) Equinox: J2000	Proper Motion RA: -7.552 mas/yr Proper Motion Dec: -18.083 mas/yr Parallax: 0.0022115" Epoch of Position: 2000	

Comments: Target acquisition (TA) star for NEXUS-OBS5-2314.
Coordinates are from Gaia DR3 with ID 1440829453269862400.
Alternate names are PS1 186152685356275572 and SDSS13 1237671940880269695.
Empirical properties: G = 17.007833 mag; BP = 18.191326 mag; RP = 15.939125 mag; BP - RP = 2.252201.
Inferred properties: T_eff = 3724.8 K; log_g = 4.6411 cm/s^2; [Fe/H] = -0.0172; A0 = 0.6466 mag.
Category=Star
Description=[M stars]
Extended=NO

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	TA___MACSJ0647-1045	RA: 06 47 46.9605 (101.9456687d) Dec: +70 12 5.88 (70.20163d) Equinox: J2000	Proper Motion RA: -1.159 mas/yr Proper Motion Dec: 5.938 mas/yr Parallax: 0.0013045" Epoch of Position: 2000	

Comments: Target acquisition (TA) star for MACSJ0647-1045.
Coordinates are from Gaia DR3 with ID 1112448825186650752.
Alternate names are PS1 192241019455762723 and AllWISE J064746.92+701205.9.
Empirical properties: G = 17.767685 mag; BP = 18.879730 mag; RP = 16.742647 mag; BP - RP = 2.137083.
Inferred properties: T_eff = 3553.6 K; log_g = 4.2996 cm/s^2; [Fe/H] = -1.3395; A0 = 0.5077 mag.
Category=Star
Description=[M stars]
Extended=NO

(9)	TA___GTO-WIDE-UDS14-4446	RA: 02 17 7.9499 (34.2831246d) Dec: -05 13 11.83 (-5.21995d) Equinox: J2000	Proper Motion RA: 6.440 mas/yr Proper Motion Dec: 1.740 mas/yr Parallax: 0.0014627" Epoch of Position: 2000

Comments: Target acquistion (TA) star for GTO-WIDE-UDS14-4446.
Coordinates are from Gaia DR3 with ID 2489663765487479808.
Alternate names are PS1 101730342831156499 and SDSS13 1237679253060780114.
Empirical properties: G = 17.727543 mag; BP = 18.712803 mag; RP = 16.757116 mag; BP - RP = 1.955687.
Inferred properties: T_eff = 4954.9 K; log_g = 3.7492 cm/s^2; [Fe/H] = -0.0098; A0 = 2.2997 mag.
Category=Star
Description=[K stars]
Extended=NO

Observation

Proposal 11376, Observation 3: Obs #3 [2314]

Mon Mar 16 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	NEXUS-OBS5-2314	RA: 17 54 10.0550 (268.5418958d) Dec: +65 08 19.71 (65.13881d) Equinox: J2000	Epoch of Position: 2000	

Comments: NEXUS-OBS5-2314 is a Little Red Dot (LRD) at redshift 4.422 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the North Ecliptic Pole (NEP) field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 TA___NEXUS-OBS5-2314	F1500W	FAST	4	1	1	11.1	270655

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
1	FASTR1	135	1	1	1	1	374.63		F560W

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	131	7	14	1	2	5122.724	270655

Special Requirements

Background Limited. Background no more than 20th percentile above minimum

Observation

Proposal 11376, Observation 4: Obs #4 [1045]

Mon Mar 16 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	MACSJ0647-1045	RA: 06 47 44.0174 (101.9334058d) Dec: +70 11 53.76 (70.19827d) Equinox: J2000	Epoch of Position: 2000	

Comments: MACSJ0647-1045 is a Little Red Dot (LRD) at redshift 4.528 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the MACS J0647+7015 (MACSJ0647) lensing cluster field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	8 TA___MACSJ0647-1045	F1500W	FAST	4	1	1	11.1	270655

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
1	FASTR1	135	1	1	1	1	374.63		F560W

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	155	20	40	1	2	17310.7	270655

Special Requirements

Background Limited. Background no more than 20th percentile above minimum

Observation

Proposal 11376, Observation 5: Obs #5 [4446]

Mon Mar 16 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	GTO-WIDE-UDS14-4446	RA: 02 17 4.9709 (34.2707121d) Dec: -05 13 3.62 (-5.21767d) Equinox: J2000	Epoch of Position: 2000	

Comments: GTO-WIDE-UDS14-4446 is a Little Red Dot (LRD) at redshift 4.683 with a V-shaped spectral energy distribution (SED) and broad-Balmer emission lines. This LRD is located in the CANDELS Ultra Deep Survey (UDS) field and was selected from the Dawn JWST Archive (DJA).

The quoted uncertainties on the coordinates correspond to the short-wavelength pixel scale for JWST/NIRCam.
Category=Galaxy
Description=[Active galaxies, Compact galaxies, High-redshift galaxies]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	9 TA___GTO-WIDE-UDS14-4446	F560W	FAST	6	1	1	16.65	270021

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
1	FASTR1	135	1	1	1	1	374.63		F560W

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	135	9	18	1	2	6787.748	270021

Special Requirements

Background Limited. Background no more than 20th percentile above minimum